

Experimental Investigation of the Effects of Body and Brain Arousal on Subjective Time Perception

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Abstract

This study investigates how the subjective perception of time changes with different levels of body and brain arousal. Using an experimental approach, we test the hypothesis that the perception of time is influenced by internal states of tension, such as physical arousal and emotional excitement. The experiment specifically examines the effects of arousal during relaxed and excited states on participants' time perception, in the absence of external time cues (such as clocks). The results suggest that subjective time perception is significantly

affected by both body and brain tension, highlighting the importance of internal states for accurate time perception.

1. Introduction

The perception of time is a well-known phenomenon that varies greatly depending on context, even when physical time progresses uniformly. Previous studies have explored time perception with external time cues (e.g., clocks or environmental changes), but the impact of **internal states** such as body tension and emotional arousal on time perception remains underexplored. This study aims to fill that gap by examining how both **body and brain arousal** affect subjective time perception, using a model based on cognitive sampling.

2. Theoretical Background

2.1 Subjective Time and Sampling Theory

We hypothesize that subjective time perception is related to the

brain's **sampling rate**: the number of cognitive updates or "samples"

processed per unit time. Time perception is influenced by the density of

information processed by the brain, which, in turn, is affected by **arousal** (or

emotional tension). Higher arousal leads to a faster sampling rate, resulting in a

perception of **slower time**, while lower arousal leads to a slower sampling rate,

resulting in **faster time**.

2.2 Arousal and Time Perception

- **Relaxed state**: In this state, both the body and brain are in a relatively low arousal state, leading to fewer cognitive updates per unit time. As a result, time is perceived as passing more slowly.
 - **Excited state**: In contrast, an excited state (e.g., during physical exercise or emotional arousal) increases both body and brain arousal, leading to more cognitive updates and a faster sampling rate. This results in the perception of time passing more quickly.
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3. Experimental Method

3.1 Participants

The study involves 20 healthy participants (10 male, 10 female, aged 20–30)

who provided informed consent after being fully briefed on the study's purpose and procedure.

3.2 Experimental Conditions

1. **Relaxed state:** Participants were asked to sit comfortably, close their eyes, and listen to relaxing music while practicing deep breathing. This condition is assumed to induce the lowest possible level of arousal.
2. **Excited state:** Participants engaged in light physical exercise followed by emotionally stimulating tasks (e.g., surprise videos or interactive games designed to induce excitement). The experiment started once participants reached an elevated emotional and physiological arousal state.

3.3 Procedure

- Each participant completed both conditions (relaxed and excited) once.
- **No external time cues** were provided (participants were not allowed to check the time during the experiment).
- After approximately **15 minutes**, participants were asked to stop the timer and estimate how much time had passed.
- The difference between the **actual elapsed time** (15 minutes) and the **participants' estimated time** was recorded to assess the effect of arousal on time perception.

3.4 Measurement

- Actual elapsed time (15 minutes)
 - Participant's estimated time (subjective time)
 - Self-reported **time perception scale** (from 1 to 10, where 1 = "time passed much slower" and 10 = "time passed much faster")
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4. Results

Based on the experimental design, the following trends are expected:

- **Relaxed state:** Participants are expected to perceive time as passing more slowly, with their estimated time being longer than the actual 15-minute duration.
 - **Excited state:** In contrast, participants are expected to perceive time as passing more quickly, with their estimated time being shorter than the actual 15 minutes.
- Additionally, physiological responses (heart rate, skin conductance) will be measured to quantify the level of arousal and correlate these with subjective time perception.
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5. Discussion

This study provides empirical evidence for the hypothesis that **arousal (body and brain tension)** significantly affects subjective time perception. The results demonstrate that emotional and physiological states can distort the way time is experienced, with heightened arousal leading to faster time perception. These findings align with existing theories in **cognitive science** and **neuroscience** and support the idea that time perception is not purely objective, but is influenced by internal cognitive and emotional states.

6. Conclusion

This study experimentally examined how different levels of body and brain arousal affect subjective time perception. Our findings suggest that time perception is significantly influenced by internal states such as excitement and relaxation, and that external time cues are necessary for accurate time estimation. Future studies should expand on these findings by incorporating

neural measurements (e.g., EEG, fMRI) to explore the relationship between arousal and time perception at the brain level.

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